Wireless Digital Communications

Topic 3

Digital Modulation

Outline

- Linear Modulation
- Modulation Performance in AWGN Channels
- Modulation Performance in Fading Channels

Digital Modulation

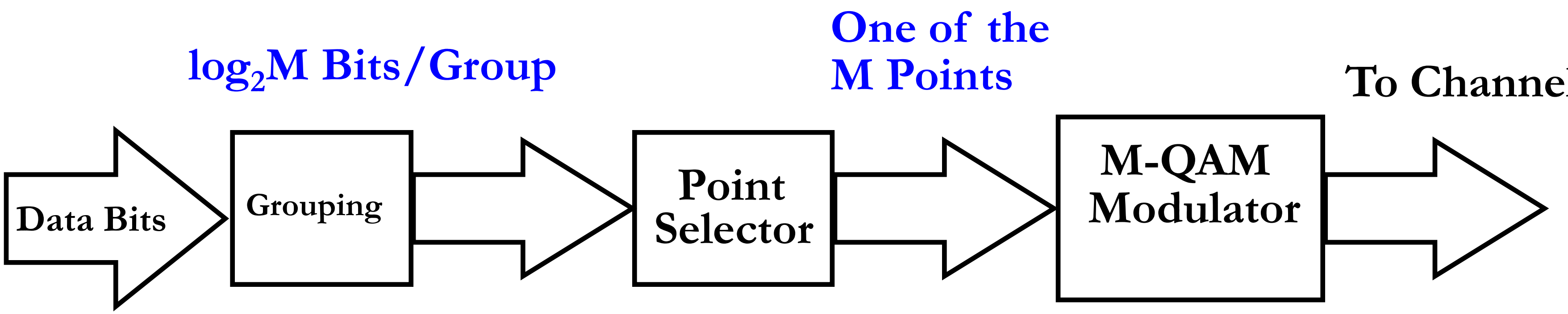
- Design Issues (Often Conflicting)
  - Data Rate
  - Spectral Efficiency
  - Power Efficiency
  - Performance (Channel Impairments and Noise)
  - Cost

Binary message sequence is divided into words of length K Bits, sent every T seconds

M possible symbols $\{m_1,\ldots m_M\}$ with probabilities $\{p_1,\ldots p_M\}$

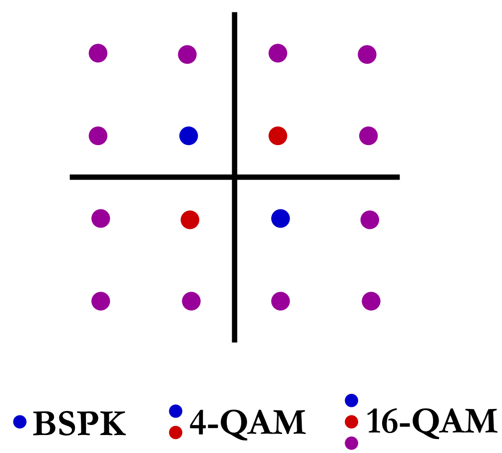
$M = 2^K$, $K = \log_2 M$, $R = K/T$ bits per second

Procedure of Modulation



$$s_i(t) = s_{i1}\sqrt{\frac{2}{T}}\cos 2\pi f_c t + s_{i2}\sqrt{\frac{2}{T}}\sin 2\pi f_c t$$

$$0 \le t \le T$$



Geometric Signal Representation

m_i is represented by signal $s_i(t)$, $0 \le t \le T$

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\phi_j(t) \quad 0 \le t \le T$$

$\{\phi_1(t), \ldots \phi_N(t)\}$ are orthonormal basis functions:

$$\int_0^T \phi_i(t)\phi_j(t)dt = \begin{cases} 1 & i = j \\ 0 & i \ne j \end{cases}$$

Linear Passband Modulation: N = 2

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos 2\pi f_c t \quad \phi_2(t) = \sqrt{\frac{2}{T}}\sin 2\pi f_c t$$

$$s_i(t) = s_{i1}\sqrt{\frac{2}{T}}\cos 2\pi f_c t + s_{i2}\sqrt{\frac{2}{T}}\sin 2\pi f_c t$$

Symbol Energy $\boxed{E_s = \int_0^T s_i^2(t)dt = s_{i1}^2 + s_{i2}^2}$

Approximate Orthogonality

$$\int_0^T \phi_1^2(t)dt = \frac{2}{T}\int_0^T .5[1+\cos(4\pi f_c t)]dt = 1 + \frac{\sin(4\pi f_c T)}{4\pi f_c T}$$

$$\int_0^T \phi_1(t)\phi_2(t)dt = \frac{2}{T}\int_0^T .5\sin(4\pi f_c t)dt = \frac{-\cos(4\pi f_c T)}{4\pi f_c T} \approx 0$$

where the approximation is taken as an equality for $f_c T >> 1$.

Geometric Signal Representation

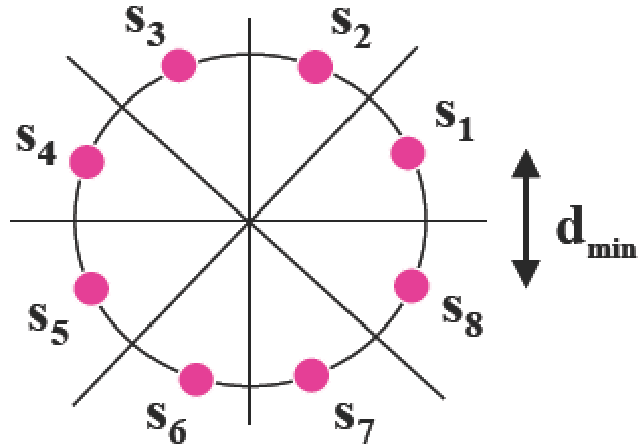
Want to minimize $P_e = p(\text{decode } m_j | m_i \text{ sent})$

Vector space analysis

$s_i(t)$ is characterized by vector $\mathbf{s}_i$

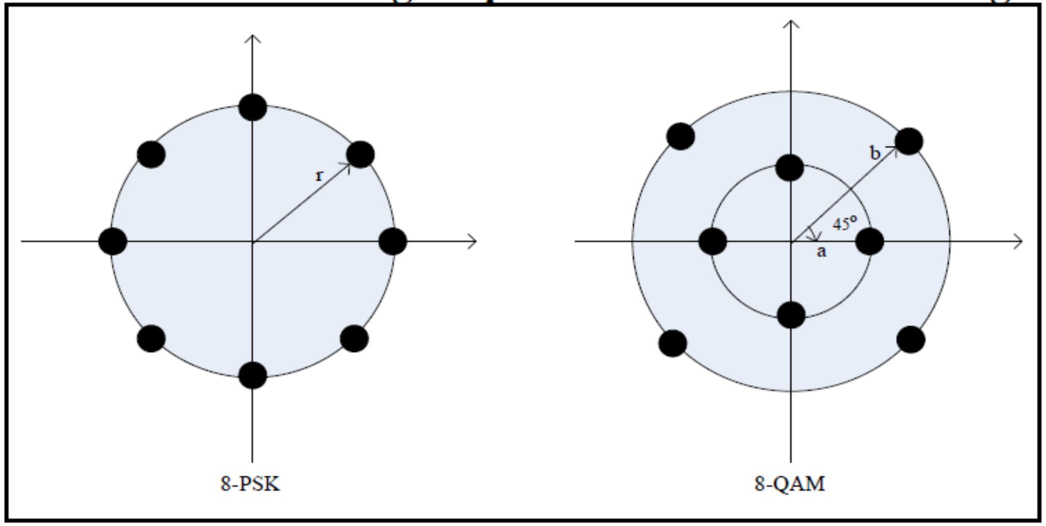
$\mathbf{s}_i = (s_{i1}, s_{i2}, \ldots, s_{iN})$

Signal Constellation:



Example (T=1)

Consider the octal signal point constellation in the figure shown below



(a) The nearest neighbor signal points in the 8-QAM signal constellation are separated in distance by A. Determine the radii a and b of the inner and outer circles.
(b) The adjacent signal points in the 8-PSK are separated by a distance of A. Determine the radius r of the circle.
(c) Determine the average transmitter powers for the two signal constellations and compare the two powers.

Example

Solution:

(1) $a^2=(\frac{A}{2})^2+(\frac{A}{2})^2=\frac{A^2}{2}$, $a=\frac{1}{\sqrt{2}}A=0.7071A$

$c^2=A^2-(\frac{A}{2})^2=\frac{3}{4}A^2$

$b=\frac{A}{2}+c=\frac{1+\sqrt{3}}{2}A=1.366A$

(2) $\frac{\frac{A}{2}}{r}=\sin\frac{\pi}{8}$, so $r=\frac{A}{2\sin\frac{\pi}{8}}=1.3066A$

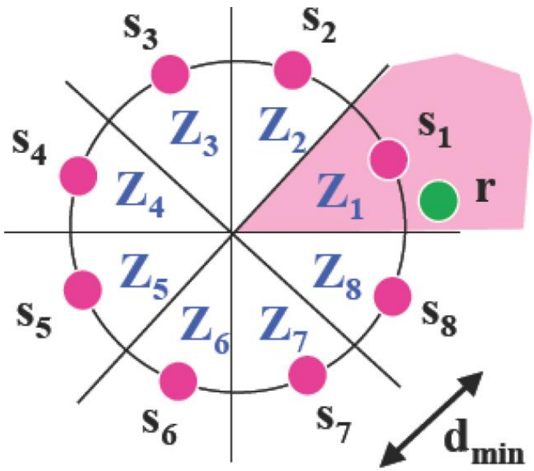
(3) 8PSK: $P=r^2=1.7071A^2$

8QAM: $P=\frac{4a^2+4b^2}{8}=\frac{3+\sqrt{3}}{4}A^2=1.1830A^2$

The Optimum Decision Rules

Maximum a posteriori probability (MAP) rule:

$$\max_{\mathbf{s}_i} p(\mathbf{s}_i\, sent \mid \mathbf{r}\, received)$$



By Bayes' rule,

$$\max_{\mathbf{s}_i} \frac{p(\mathbf{r} \mid \mathbf{s}_i)\, p(\mathbf{s}_i)}{p(\mathbf{r})} = \max_{\mathbf{s}_i} p(\mathbf{r} \mid \mathbf{s}_i)\, p(\mathbf{s}_i)$$

The Optimum Decision Rules

$$(p(\mathbf{s}_i) = 1/M)$$

$$= \max_{\mathbf{s}_i} p(\mathbf{r}|\mathbf{s}_i \, sent)$$

Maximum likelihood (ML) rule

Conditional Distribution （Likelihood function）

$$p(\mathbf{r}|\mathbf{s}_i \text{ sent}) = \prod_{j=1}^{N} p(r_j|m_i) = \frac{1}{(\pi N_0)^{N/2}} \exp\left[-\frac{1}{N_0}\sum_{j=1}^{N}(r_j - s_{ij})^2\right]$$

$$L(\mathbf{s}_i) = p(\mathbf{r}|\mathbf{s}_i\, sent)$$

log likelihood function:

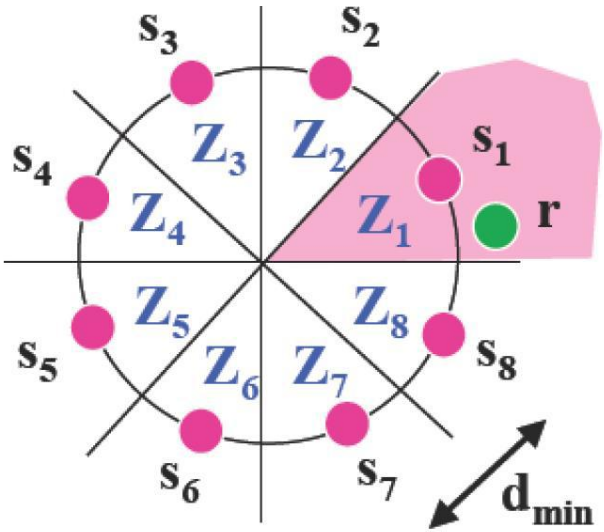
$$l(\mathbf{s}_i) = -\frac{1}{N_0}\sum_{j=1}^{N}(r_j - s_{ij})^2 = -\frac{1}{N_0}\left\|\mathbf{r} - \mathbf{s}_i\right\|^2$$

MEP➡MAP➡ ML➡Minimum distance rule

Decision Regions

- Optimum receiver (minimum probability of error) is based on Maximum Likelihood (ML) estimation

- ML receiver decodes s_i closest to $\mathbf{r}$ (observation vector)



Decision Regions-Error Probability

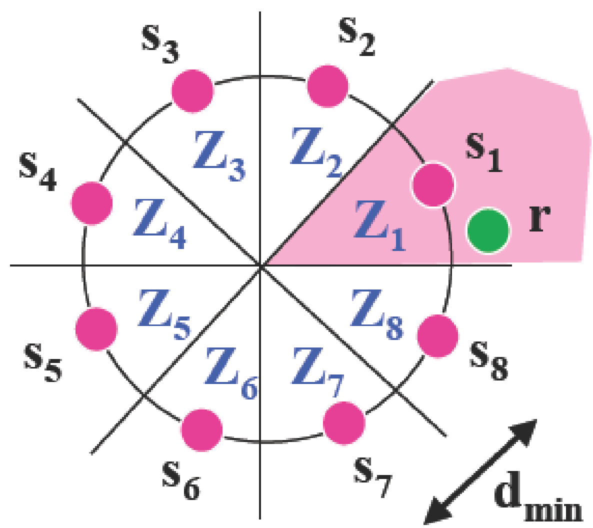
Assign decision regions:

$$Z_i = \{r : p(s_i \text{ sent} | r) > p(s_j \text{ sent} | r \text{ for all } j \neq i\}$$

$$= \{r : |x\text{-}s_i| < |x\text{-}s_j| \text{ all } j \neq i\}$$

$$r \in Z_i \Rightarrow m = m_i$$

Signal Constellation

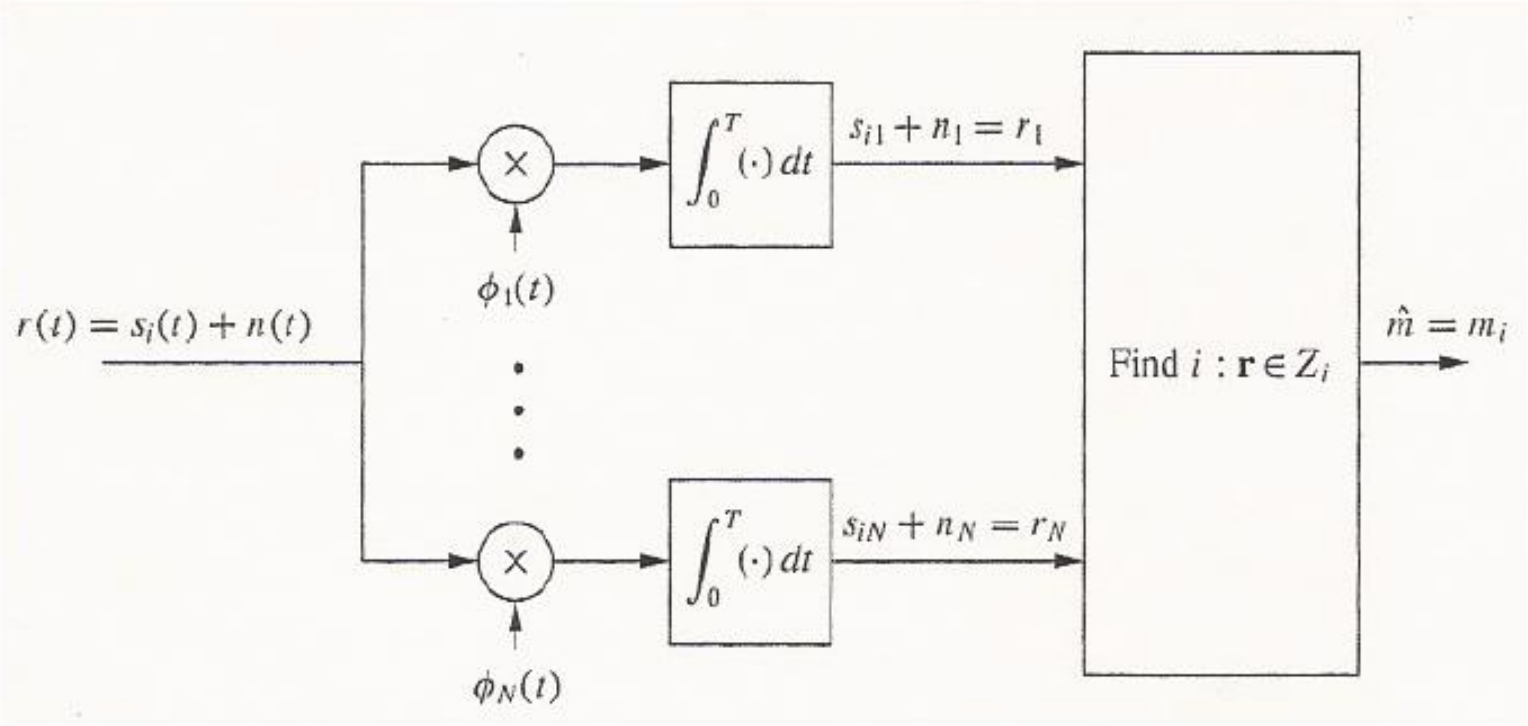


P_e is based on noise distribution

$$P_s \leq (M-1)Q\left(\sqrt{d_{\min}^2/(2N_0)}\right)$$

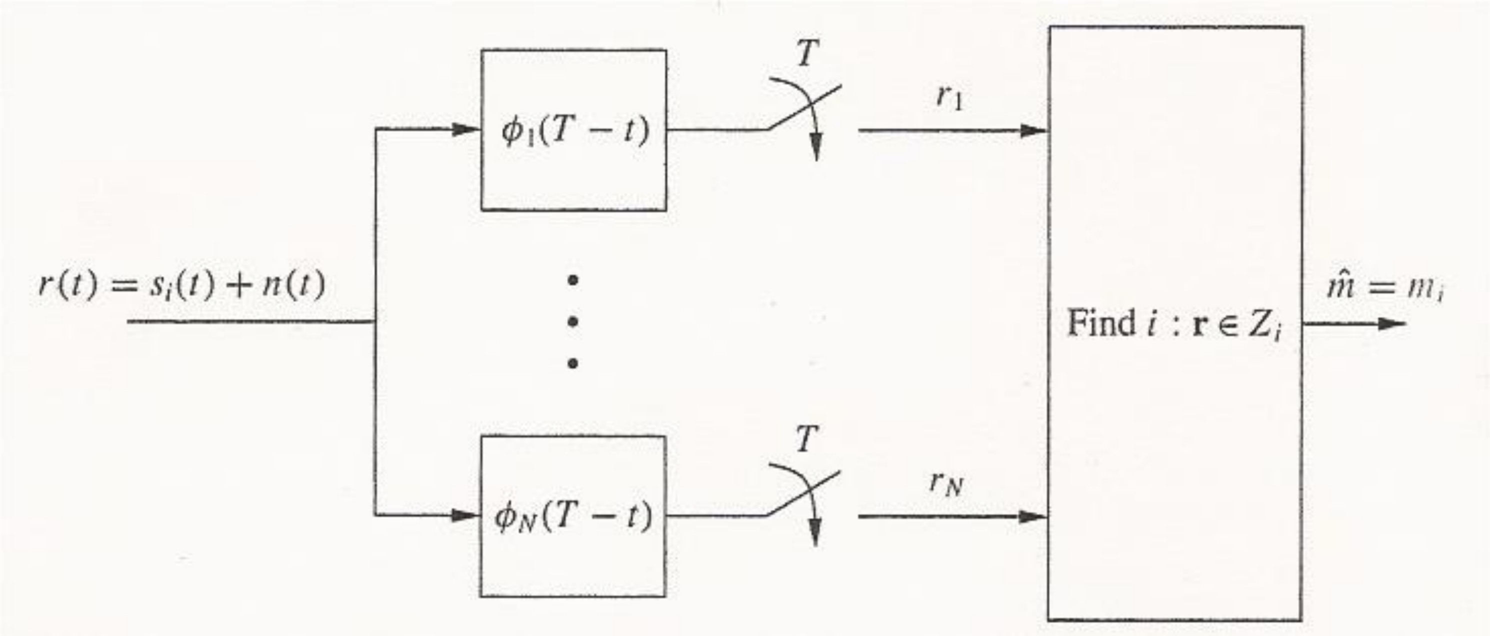
Optimum Coherent Detection
Correlation Receiver

Received signal (transmitted signal plus Gaussian noise) is correlated with each basis function.



Optimum Coherent Detection Matched Filter Receiver

Received signal (transmitted signal plus White Gaussian noise) is passed through a bank of filters, each matched to one basis function.



The Equivalence

$$h_j(t) = \phi_j(T - t)$$

$$y_j(t) = \int_{-\infty}^{\infty} x(\tau) h_j(t - \tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) \phi_j(T - t + \tau) d\tau$$

$$y_j(T) = \int_{-\infty}^{\infty} x(\tau) \phi_j(\tau) d\tau$$

$$= \int_{0}^{T} x(\tau) \phi_j(\tau) d\tau$$

Therefore, the correlator equals sampling at time T after a matched filter.

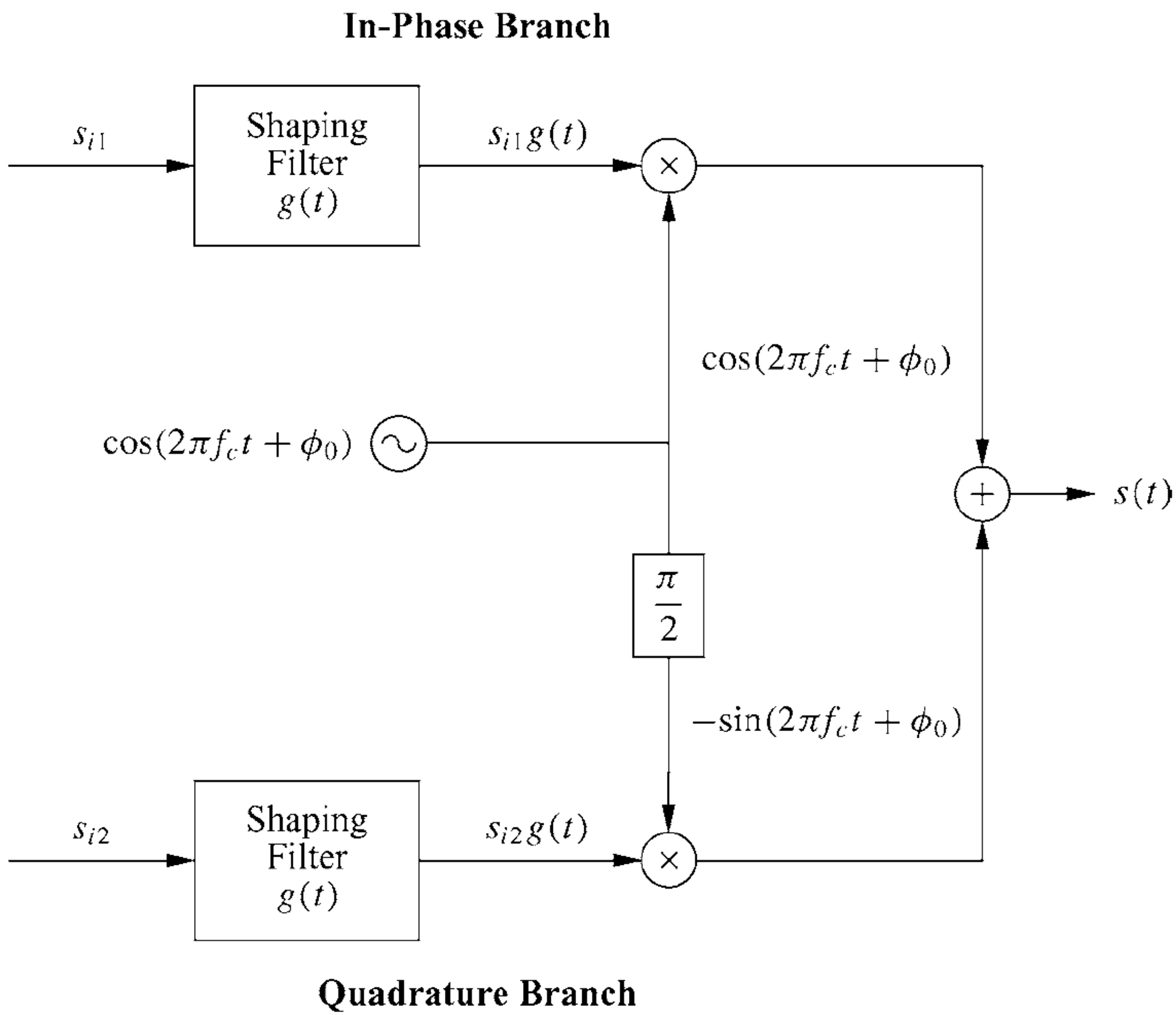
Linear Modulation

- Bits encoded in carrier amplitude or phase

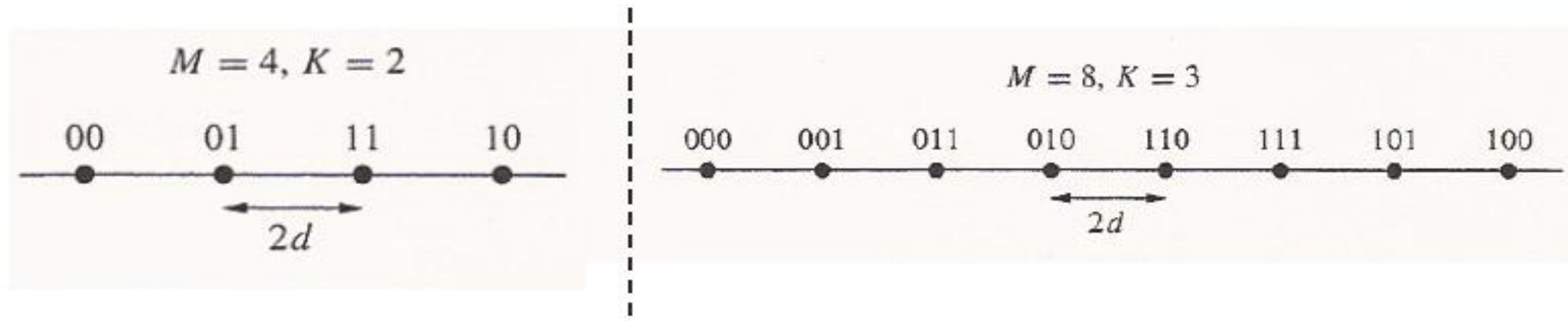
$$s(t)=\sum_n a_n g(t-nT_s)\cos(2\pi f_c t)-\sum_n b_n g(t-nT_s)\sin(2\pi f_c t)$$

  - Pulse shape g(t) typically Nyquist
  - Signal constellation defined by (a_n,b_n) pairs
  - Can be differentially encoded
  - M values for $(a_n,b_n) \Rightarrow \log_2 M$ bits per symbol
- P_s depends on
  - Minimum distance d_{min} *(depends on γ_s)*
  - # of nearest neighbors α_M
  - Approximate expression: $\boxed{P_s \approx \alpha_M Q\left(\sqrt{\beta_M \gamma_s}\right)}$

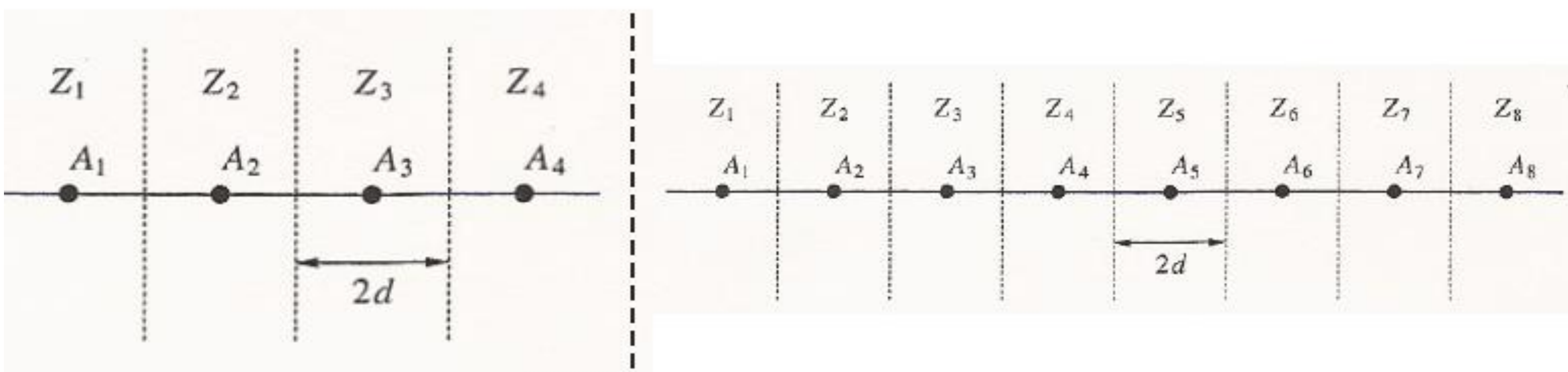
Modulator for MQAM/MPSK



Signal Constellation and Decision Regions-MPAM

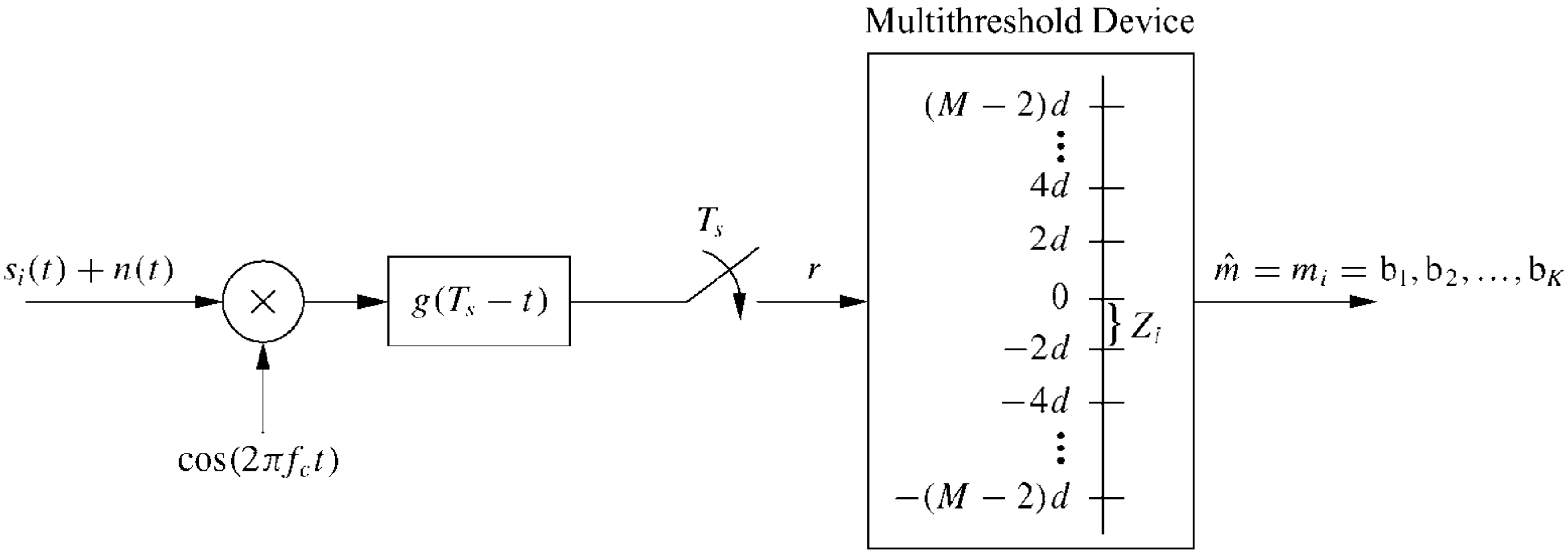


Bit Mapping by Gray Encoding

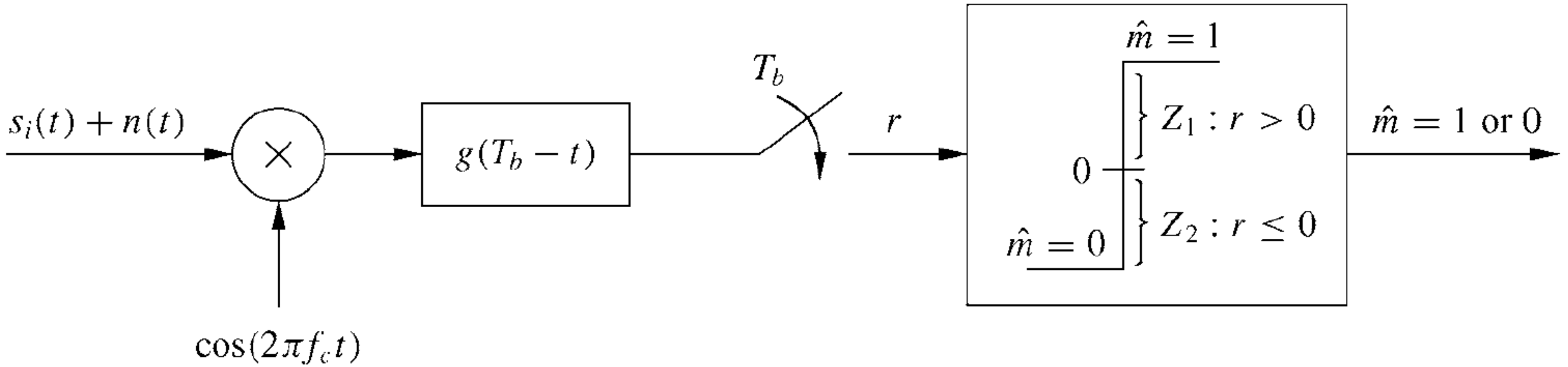


Decision Regions

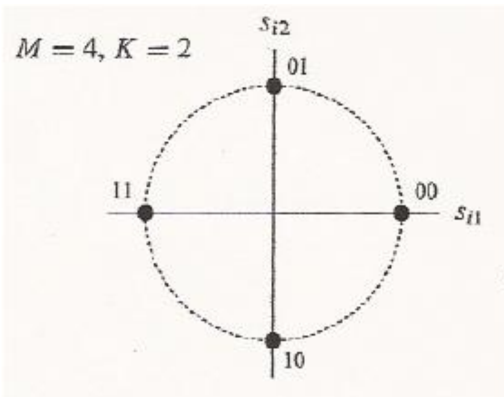
Coherent Demodulation-MPAM



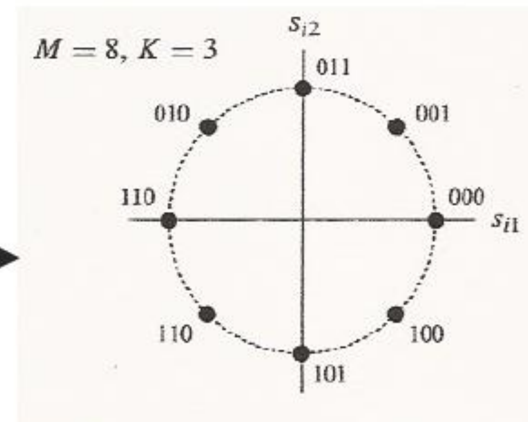
Coherent Demodulation-BPSK

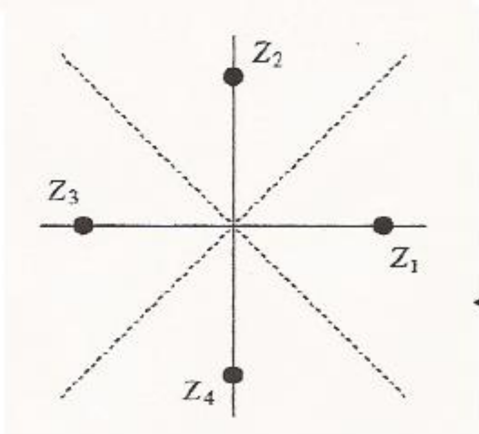


Signal Constellation and Decision Regions-MPSK

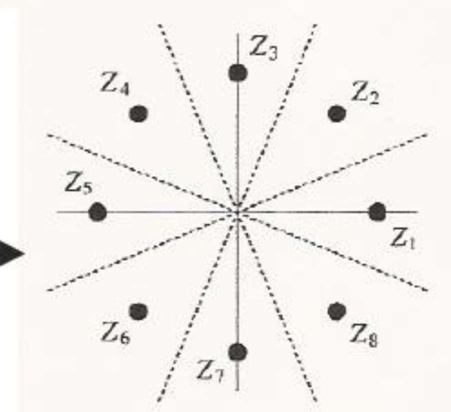


Bit Mapping by Gray Encoding

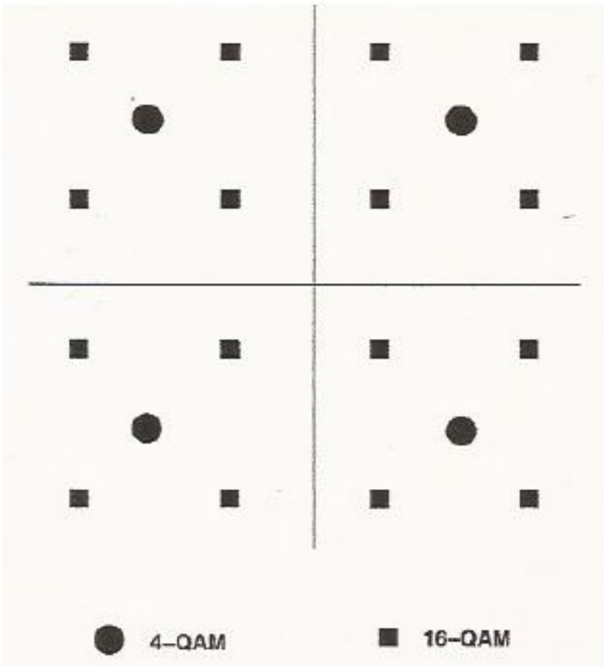




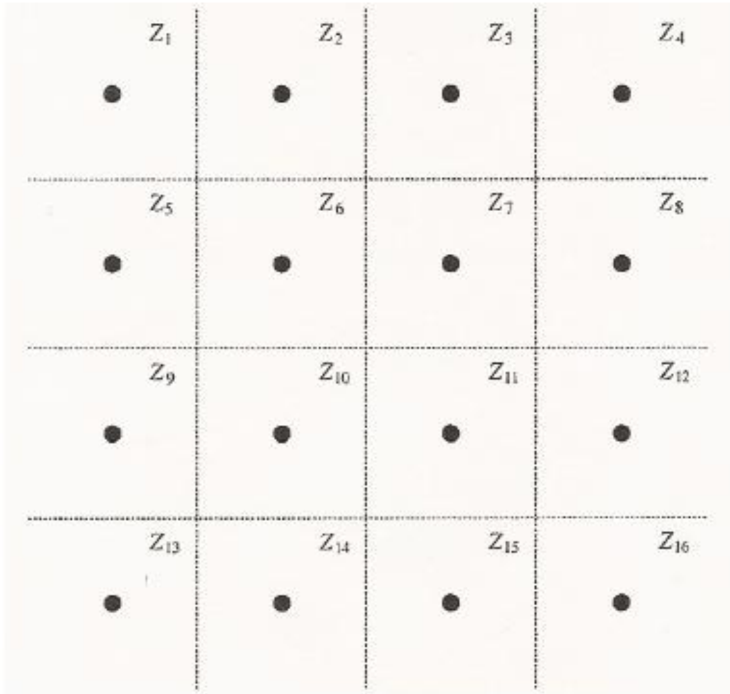
Decision Regions



Signal Constellation and Decision Regions-MQAM

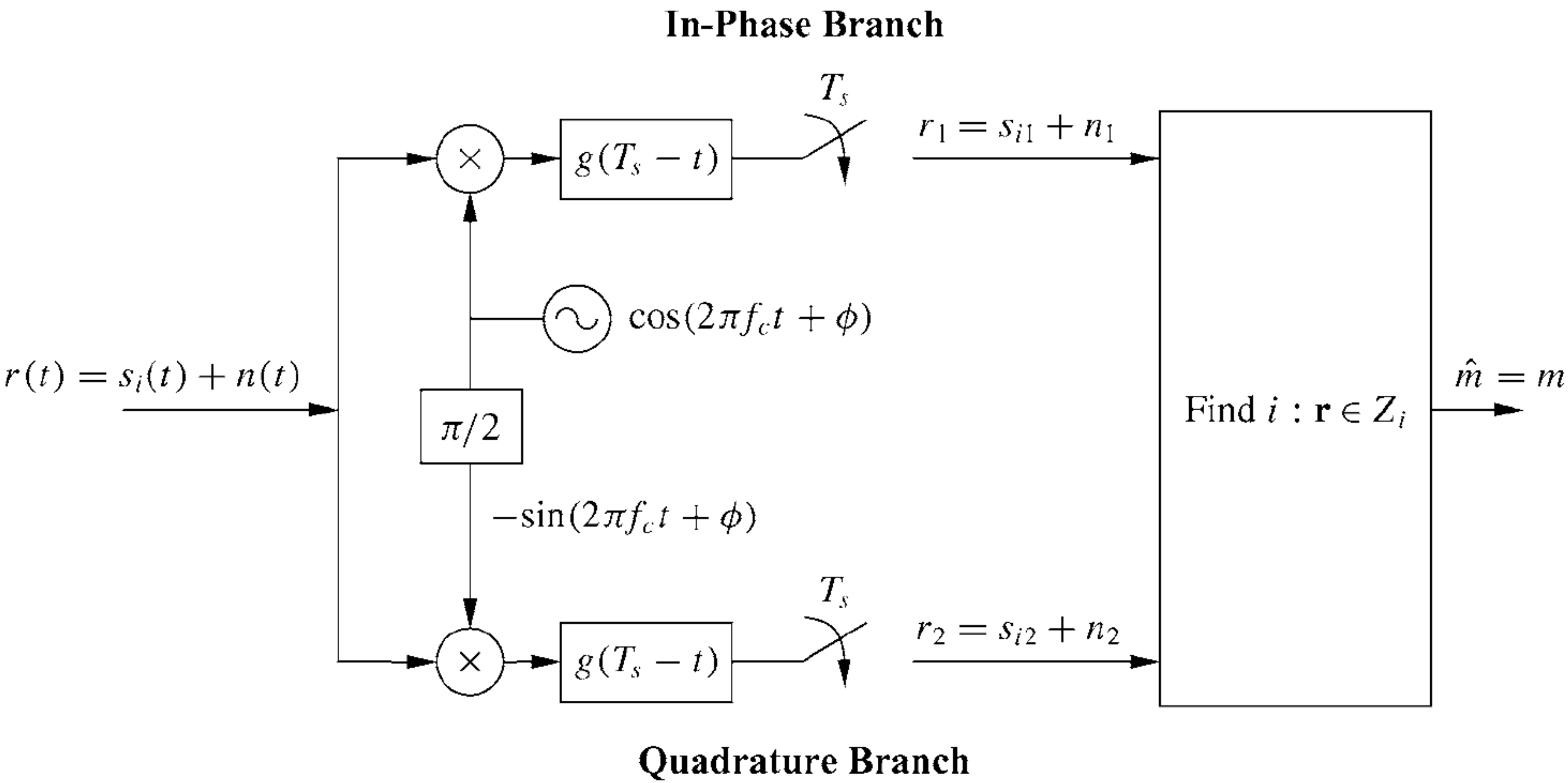


Signal Constellation



Decision Regions

Coherent Demodulation-MQAM/MPSK



Pulse Shaping

Linear Passband Modulation: N = 2

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos 2\pi f_c t \quad \phi_2(t) = \sqrt{\frac{2}{T}}\sin 2\pi f_c t$$

$$s_i(t) = s_{i1}\sqrt{\frac{2}{T}}\cos 2\pi f_c t + s_{i2}\sqrt{\frac{2}{T}}\sin 2\pi f_c t$$

Pulse shaping improves poor spectral characteristics of rectangular pulses

$$s_i(t) = s_{i1} g(t)\cos 2\pi f_c t + s_{i2} g(t)\sin 2\pi f_c t$$

Pulse Shaping

Pulse shaping function g(t) should improve spectral efficiency without compromising Zero-ISI requirement. Transmitted pulses pass through transmitter and receiver filters, and the channel. Received pulses $p_r(t)$ should satisfy the Zero-ISI condition:

Time domain condition:

$$p_r(kT_S) = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases}$$

Frequency domain condition (Nyquist Criterion):

$$\frac{1}{T_S} = \sum_{k=-\infty}^{\infty} P_r\left(f + \frac{k}{T_S}\right) = 1$$

Example: Raised Cosine Pulse Shaping

Example

速率为2400bit/s的数字信号经过4PSK调制解调器(modem)送往电话信道传输，该信道可利用的最大频率范围是500～2900Hz，发送滤波器采用升余弦滤波器，当发送信号占用信道可利用的最大频率范围时，试求：

(1) 符号速率；

(2) 载波频率；

(3) 升余弦滤波器的滚降系数。

Example

已知$r_b = 2400bps$，4PSK，B=2900-500=2400 Hz

(1) $r_s = \dfrac{r_b}{k} = \dfrac{2400}{2}$ =1200Baud

(2) $f_c = \dfrac{2900-500}{2} + 500$ =1700Hz

(3)频谱效率$\eta = r_s / B = 1/(1+\alpha)$

从中可以解出升余弦滤波器的滚降系数$\alpha = 1$

Probability of Error Analysis AWGN-Non-Fading-Coherent

T_S = Symbol Time
T_b = Bit Time

E_S = Signal Energy per Symbol
E_b = Signal Energy per Bit

Probability of error is a function of SNR

$$SNR = \frac{P_r}{N_0 B} = \frac{E_S}{N_0 B T_S} = \frac{E_b}{N_0 B T_b}$$

For pulse shaping with $T_S = 1/B$:

$$\gamma_S = \frac{E_S}{N_0}$$ **For M-ary**

$$\gamma_b = \frac{E_b}{N_0}$$ **For Binary**

Using Gray Coding and assuming errors only between neighboring symbols leads to one bit error for each symbol error

Probability of Error Analysis AWGN-Non-Fading-Coherent

M-ary versus Binary : $\gamma_b = \dfrac{\gamma_s}{\log_2 M}$ $\qquad P_b \approx \dfrac{P_s}{\log_2 M}$

Table 6.1: Approximate symbol and bit error probabilities for coherent modulations

Modulation	$P_s(\gamma_s)$	$P_b(\gamma_b)$
BFSK		$P_b = Q\left(\sqrt{\gamma_b}\right)$
BPSK		$P_b = Q\left(\sqrt{2\gamma_b}\right)$
QPSK, 4-QAM	$P_s \approx 2Q\left(\sqrt{\gamma_s}\right)$	$P_b \approx Q\left(\sqrt{2\gamma_b}\right)$
MPAM	$P_s = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\bar{\gamma}_s}{M^2-1}}\right)$	$P_b \approx \frac{2(M-1)}{M\log_2 M} Q\left(\sqrt{\frac{6\bar{\gamma}_b \log_2 M}{M^2-1}}\right)$
MPSK	$P_s \approx 2Q\left(\sqrt{2\gamma_s}\sin\left(\frac{\pi}{M}\right)\right)$	$P_b \approx \frac{2}{\log_2 M} Q\left(\sqrt{2\gamma_b \log_2 M}\sin\left(\frac{\pi}{M}\right)\right)$
Rectangular MQAM	$P_s \approx 4Q\left(\sqrt{\frac{3\bar{\gamma}_s}{M-1}}\right)$	$P_b \approx \frac{4}{\log_2 M} Q\left(\sqrt{\frac{3\bar{\gamma}_b \log_2 M}{M-1}}\right)$
Nonrectangular MQAM	$P_s \approx 4Q\left(\sqrt{\frac{3\bar{\gamma}_s}{M-1}}\right)$	$P_b \approx \frac{4}{\log_2 M} Q\left(\sqrt{\frac{3\bar{\gamma}_b \log_2 M}{M-1}}\right)$

Error Probability Approximation Coherent Demodulation

For all coherent demodulation probability of error has a functional form:

$$P_s \approx \alpha_M Q\left(\sqrt{\beta_M \gamma_s}\right)$$

where: α_m is the number of nearest neighbors in the constellation
β_m is a constant which depends on the specific modulation

$$Q(z) = \int_z^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

An alternate but simpler Q-Function suitable for AWGN and fading channels is:

$$Q(z) = \frac{1}{\pi} \int_0^{\pi/2} \exp\left[-\frac{z^2}{2\sin^2\phi}\right] d\phi \quad z > 0$$

Main Points

- Major Linear Modulation Schemes are MPAM, MPSK, MQAM
- Linear modulation more spectrally efficient but less robust than nonlinear modulation
- Decision regions are based on Maximum Likelihood
  - Optimum coherent detection structure is based on Correlation Receivers and Matched Filter Receivers
  - P_s depends on constellation minimum distance
  - P_s in AWGN approximated by: $P_s \approx \alpha_M Q\left(\sqrt{\beta_M \gamma_s}\right)$
  - Pulse-shaping improves spectral characteristics
  - An alternate Q-function is more suitable for error analysis

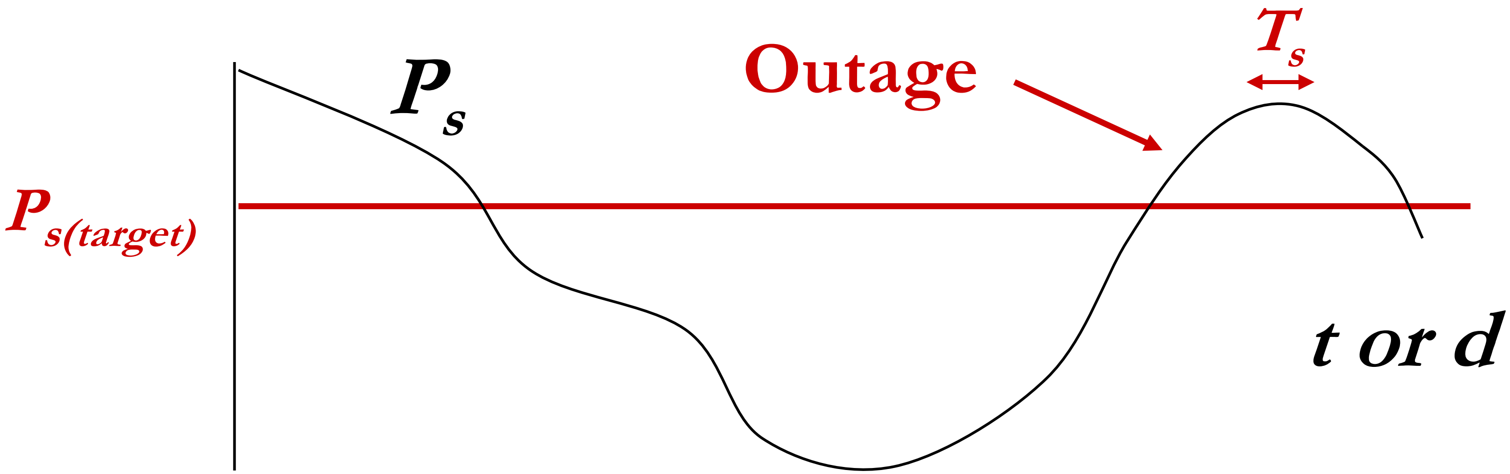
Linear Modulation in Fading

- In fading γ_s and therefore P_s random
- Performance metrics:
  - Outage probability: $p(P_s > P_{target}) = p(\gamma < \gamma_{target})$
  - Average P_s , $\overline{P}_s$:

$$\overline{P}_s = \int_0^\infty P_s(\gamma)\,p(\gamma)\,d\gamma$$

  - Combined outage and average P_s

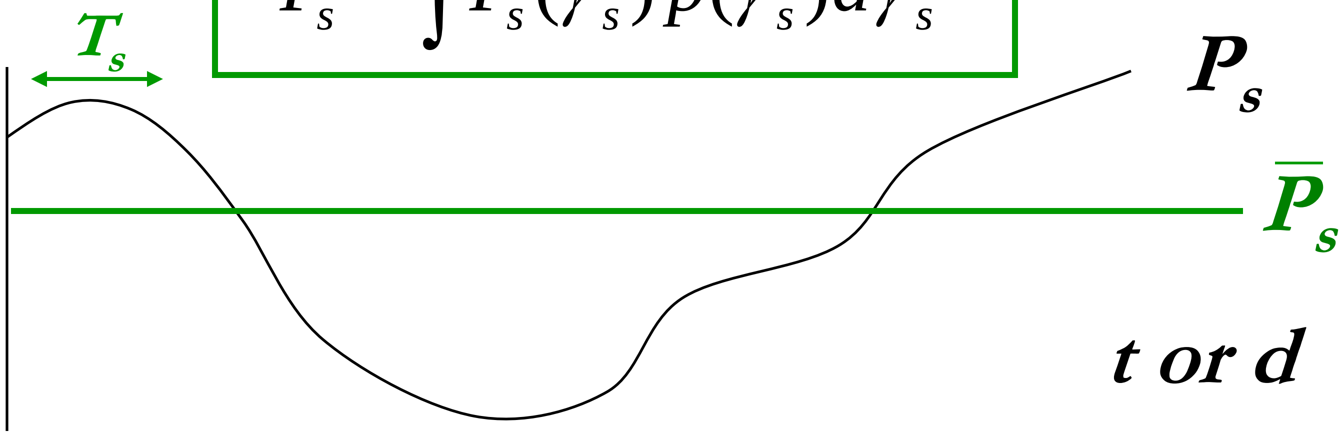
Outage Probability



- Probability that P_s is above target
- Equivalently, probability γ_s below target
- Used when $T_c >> T_s$

Average P_s

$$\overline{P}_s = \int P_s(\gamma_s)\,p(\gamma_s)\,d\gamma_s$$



- Expected value of random variable P_s
- Used when $T_c \sim T_s$
- Error probability much higher than in AWGN alone
- Alternate Q function approach greatly simplifies calculations (switch integral order, becomes Laplace Xfm)

Average BER in AWGN and Rayleigh

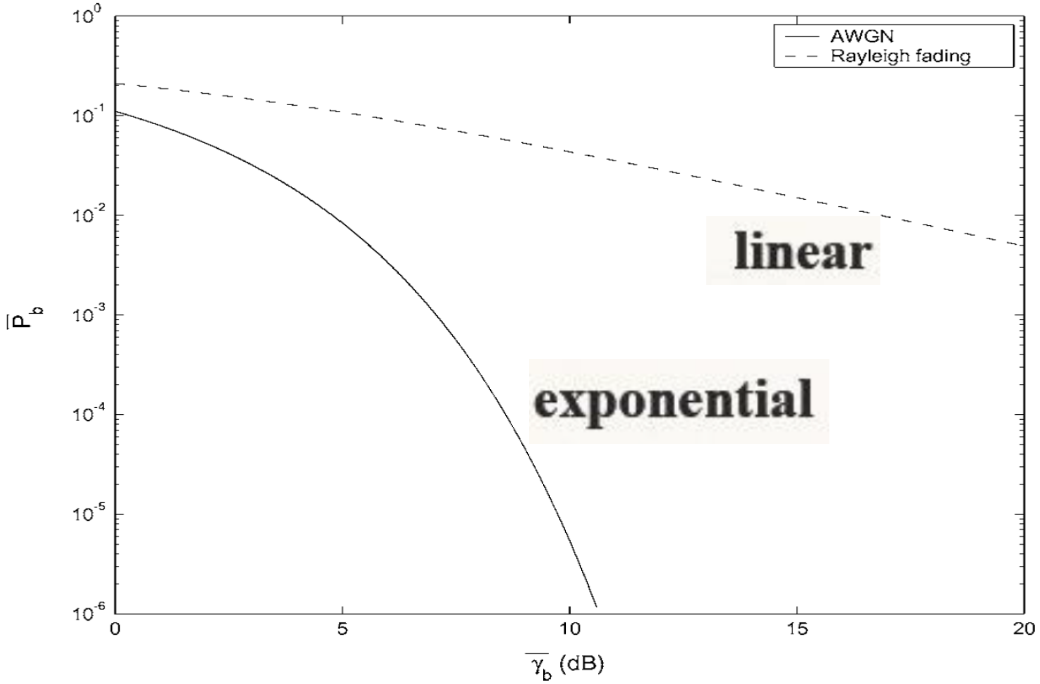


Figure 6.1: Average P_b for BPSK in Rayleigh fading and AWGN.

BER = 10^{-3} SNR = 8dB for AWGN
24dB in Rayleigh

$$BPSK: \overline{P}_b \approx 1/(4\overline{\gamma}_b) \qquad (Rayleigh \quad fading)$$

MGF Approach for Average P_s

$\bar{P}_s$ using MGF $\left(\mathcal{M}_\gamma(s) = \int_0^\infty e^{s\gamma} p(\gamma) d\gamma\right)$

$$\mathbf{E}[\gamma^n] = \frac{\partial^n}{\partial s^n} [\mathcal{M}_{\gamma_s}(s)]|_{s=0}$$

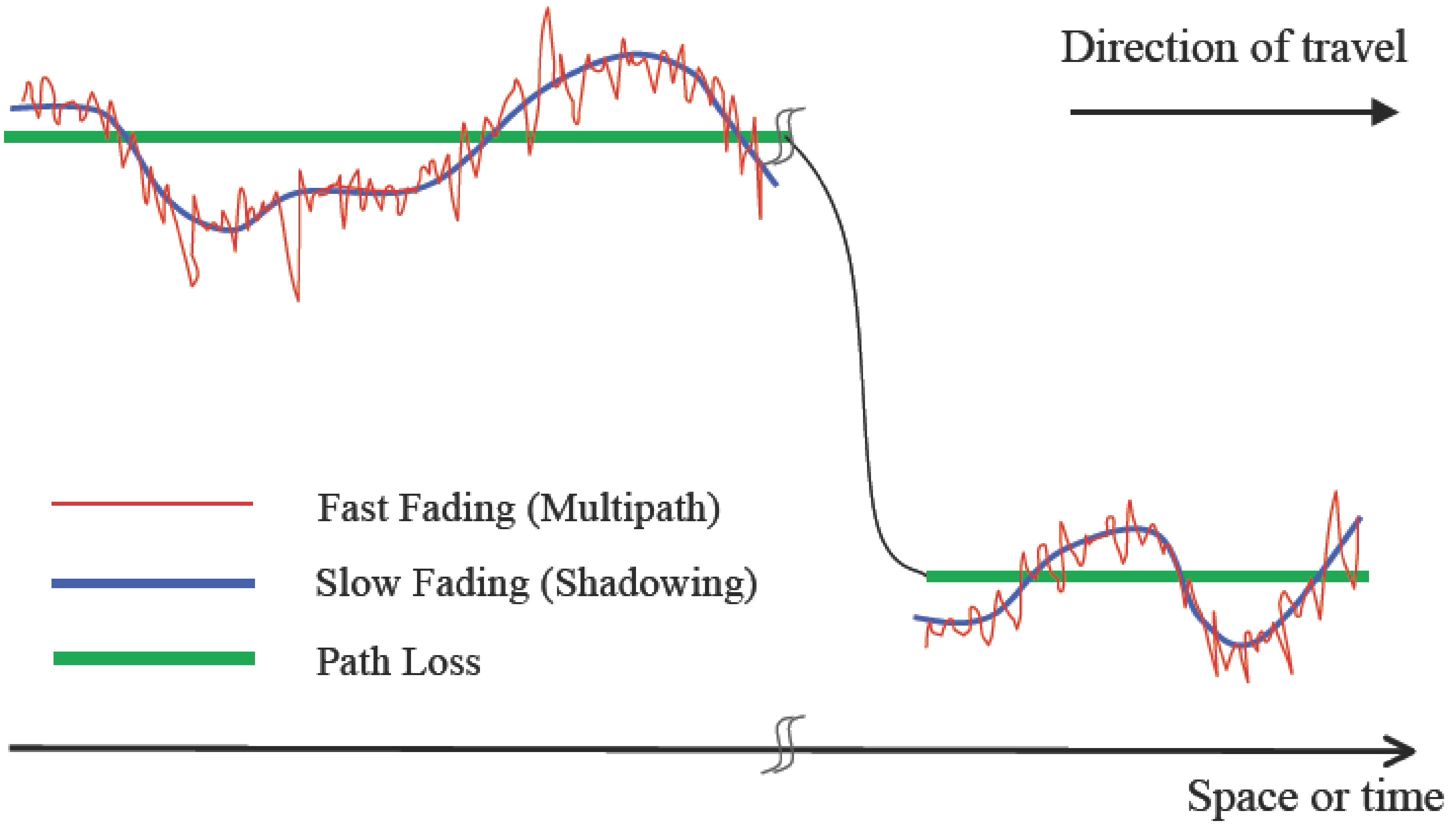
$$\overline{P}_s = \int_0^\infty \alpha_M Q(\sqrt{\beta_M \gamma}) p(\gamma) d\gamma$$

$$= \frac{\alpha_M}{\pi} \int_{\gamma=0}^{\gamma=\infty} \int_{\phi=0}^{\phi=\pi/2} e^{-\beta_M \gamma / 2\sin^2\phi} p(\gamma) d\phi d\gamma$$

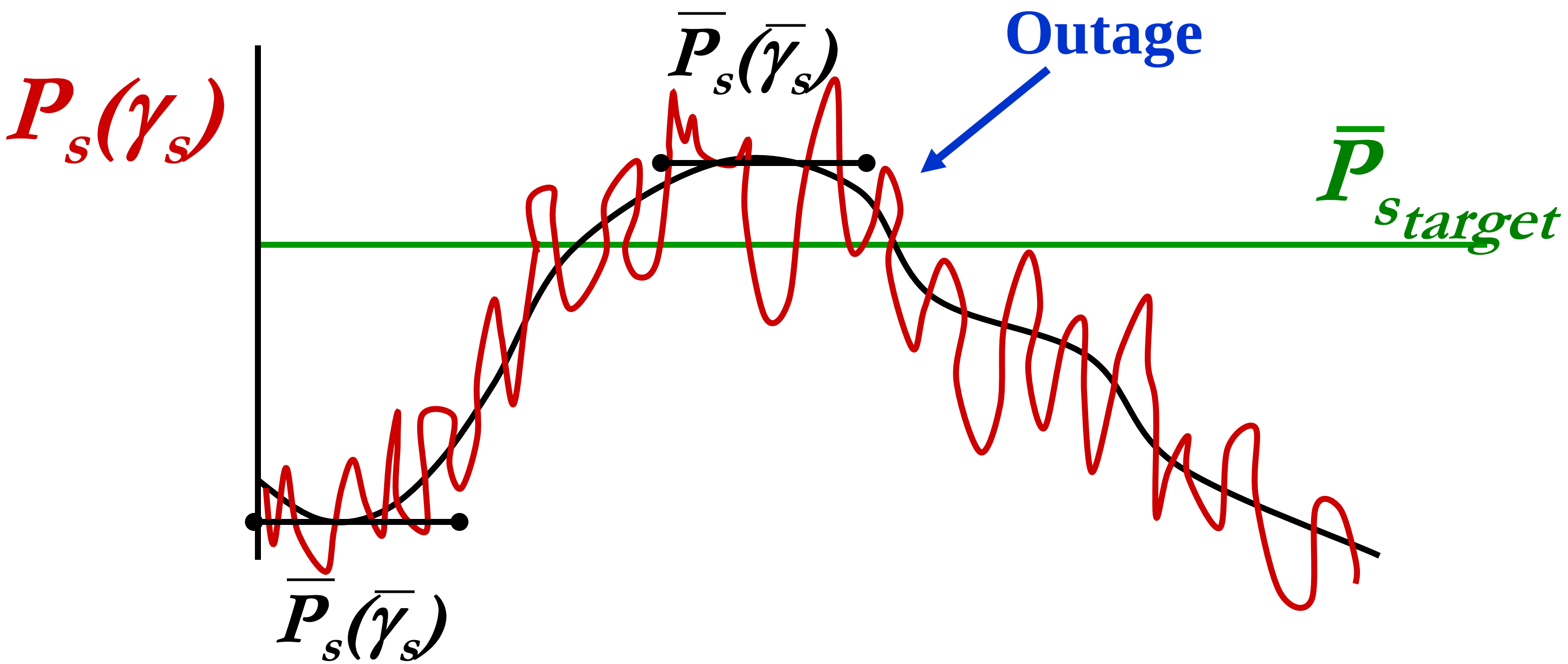
$$= \frac{\alpha_M}{\pi} \int_{\phi=0}^{\phi=\pi/2} \int_{\gamma=0}^{\gamma=\infty} e^{-\beta_M \gamma / 2\sin^2\phi} p(\gamma) d\gamma d\phi$$

$$= \frac{\alpha_M}{\pi} \int_{\phi=0}^{\phi=\pi/2} \mathcal{M}_\gamma(-\beta_M / 2\sin^2\phi) d\phi$$

Signal Variations



Combined outage and average P_s



- Used in combined shadowing and flat-fading
- $\bar{P}_s$ varies slowly, locally determined by flat fading
- Declare outage when $\bar{P}_s$ above target value

Doppler Effects

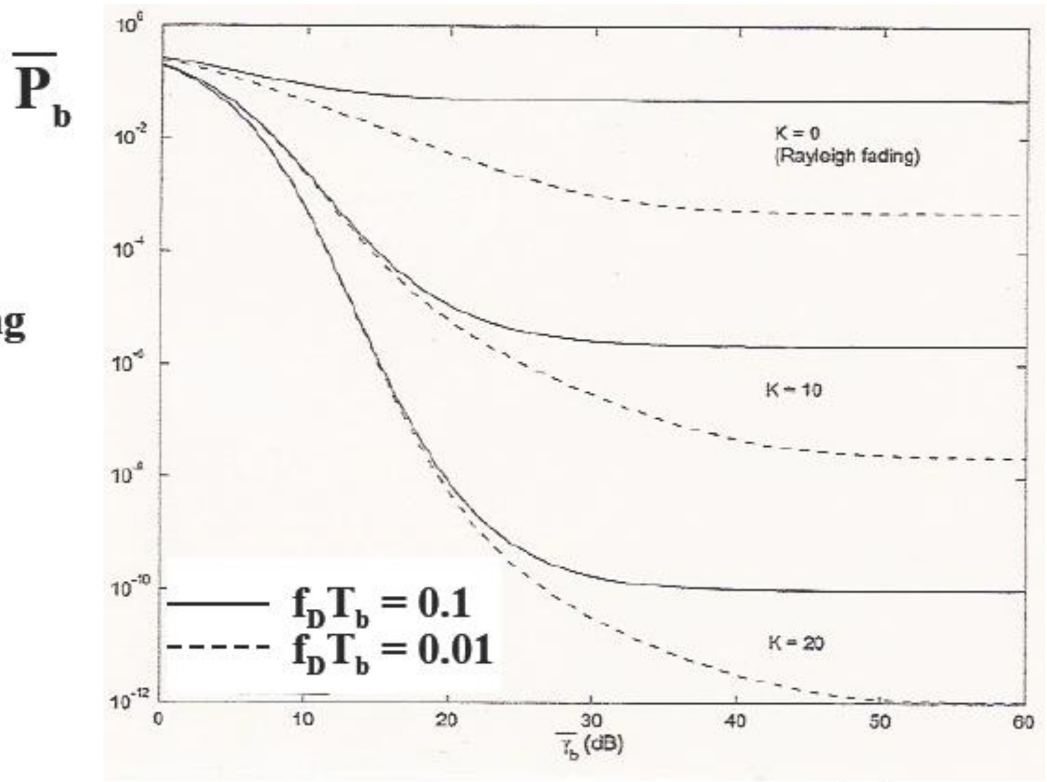
- High doppler causes channel phase to decorrelate between symbols
- Leads to an irreducible error floor for differential modulation
  - Increasing power does not reduce error
- Error floor depends on $f_D T_b$ as

$$P_{floor} = \frac{1 - J_0(2\pi f_D T_b)}{2} \approx .5(\pi f_D T_b)^2$$

Irreducible BER duo to Doppler

$f_D T_b \; \alpha \; T_b / T_c$

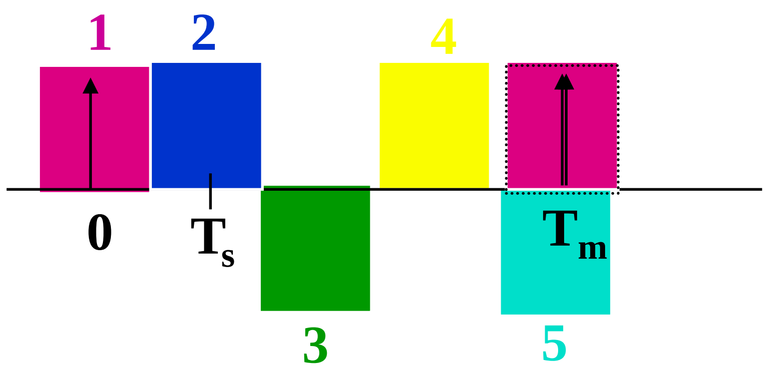
For fixed f_D decreasing T_b (increasing R_b) decreases error floor

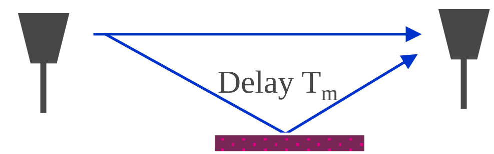


DPSK in fast Rician fading $\qquad \overline{\gamma}_b$ (dB)

Delay Spread (ISI) Effects

- Delay spread exceeding a symbol time causes ISI (self interference).

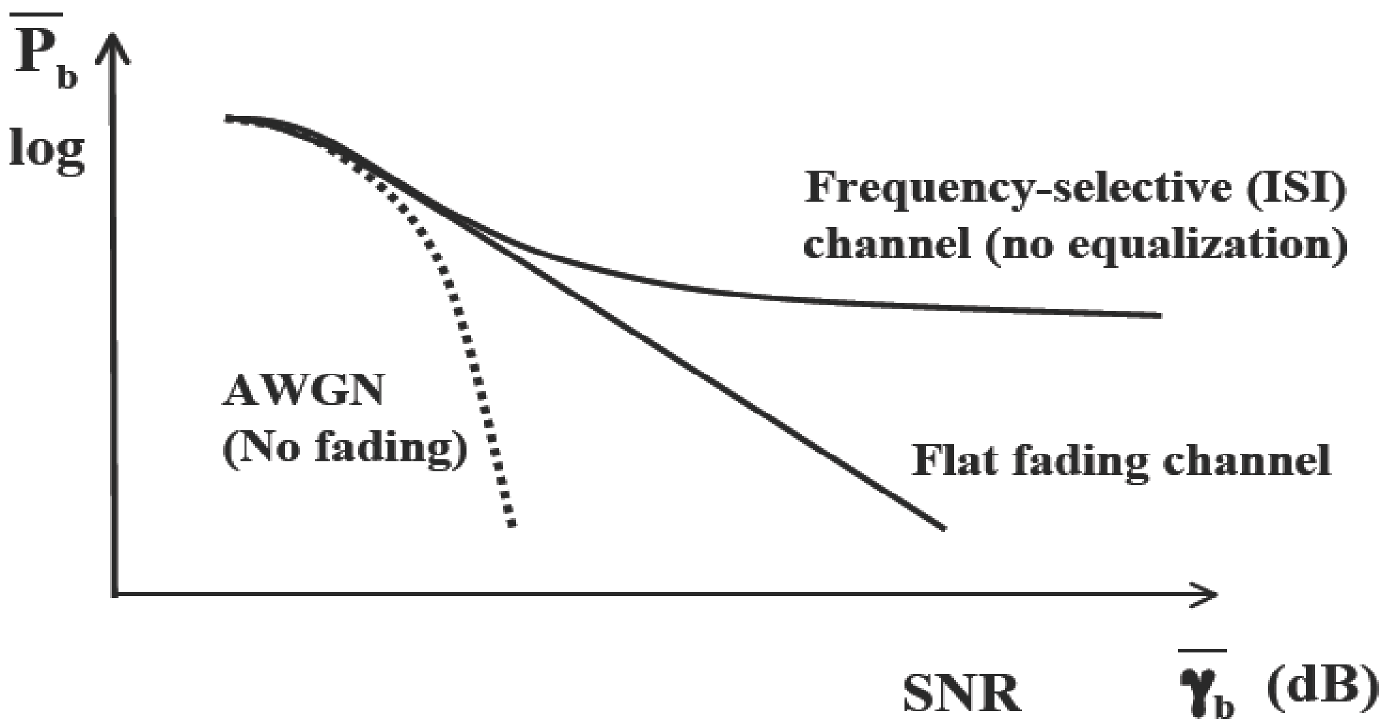




- ISI leads to irreducible error floor: $\overline{P}_{b,floor} \approx (\sigma_{T_m}/T_s)^2$
  - Increasing signal power increases ISI power
- ISI imposes data rate constraint: $T_s >> T_m$ ($R_s << B_c$)

$$R \le \log_2(M) \times \sqrt{\overline{P}_{b,floor}/\sigma^2_{T_m}}$$

Typical BER Versus SNR Curves



$$SINR = P_r / (N_0 B + ISI) \qquad SNR = P_r / (N_0 B)$$

$$P_r \uparrow \rightarrow \left\langle \begin{matrix} SNR \uparrow \\ ISI \uparrow \end{matrix} \right\rangle \rightarrow SINR \quad remains \quad unchanged$$

Modulation for LTE

- Data Channel: QPSK， 16QAM， 64QAM （adaptive）
- Control Channel: BPSK， QPSK （fixed）

Modulation for Major Standards

Second Generation

GSM:	GMSK
IS136:	π/4DQPSK
IS-95	BPSK/QPSK
PDC	π/4DQPSK

Third Generation

CDMA2000:	QPSK (DL), BPSK (UL) Phase I
	MPSK (DL), QPSK (UL) Phase II
W-CDMA:	QPSK (DL), BPSK (UL)

Modulation for Major Standards

Wireless LAN

802.11:	BPSK, QPSK
802.11a:	BPSK, QPSK, MQAM
802.11b	BPSK, QPSK
802.11g	BPSK, QPSK, MQAM

Short Range Wireless Network

ZigBee (802.15.4):	BPSK, OQPSK
Bluetooth(802.15.1):	GFSK
UWB (802.15.3) (proposal)	BPSK, QPSK

Main Points

- Fading greatly increases average P_s
  - Alternate Q function approach simplifies P_s calculation, especially its average value in fading
  - Moment Generating Function approach can be used effectively for average error probability calculations in fading
- Doppler spread only impacts differential modulation causing an irreducible error floor at low data rates
- Delay spread causes ISI and irreducible error floor or imposes limits on transmission rates
- Need to combat flat and frequency-selective fading
  - Focus of the rest of the course.

- Ex. 5.10, 5.17